

EduVision_DV (Higher Education Performance Dashboard)

Project Statement

This project aims to develop a comprehensive higher education analytics dashboard suite for analyzing university rankings, research performance, student diversity, academic excellence, and global education trends.

Using publicly available datasets such as QS World University Rankings and Times Higher Education World University Rankings, the project will transform raw educational data into actionable insights through interactive Tableau dashboards.

The dashboard suite will enable students, academic researchers, university administrators, policymakers, and education consultants to evaluate institutional performance, compare universities globally, analyze research output, and identify trends in higher education.

The final deliverable is a unified Tableau workbook consisting of four interconnected dashboards:

- University Overview
- Research Analytics
- Student Analytics
- Country Comparison

Outcomes

- Collection and integration of global university datasets.
- Data cleaning and transformation for educational analytics.
- Development of higher education KPIs.
- Creation of four interactive Tableau dashboards.
- University ranking and performance analysis.
- Country-level education benchmarking.

- Interactive filters and drill-down capabilities.
 - Single Tableau workbook (.twbx) as final deliverable.
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Modules to be Implemented

- Education data collection
 - Data cleaning and transformation
 - KPI engineering
 - Dashboard planning and wireframing
 - Dashboard development
 - Dashboard integration
 - Testing and validation
 - Documentation and project delivery
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Workflow Diagram



Milestone 1: Data Collection and Preparation (Weeks 1–2)

Module 1: University Data Collection

Tasks

- Download QS Ranking datasets.
- Download World University Ranking datasets.
- Collect university performance indicators.
- Merge ranking datasets into a common structure.

Deliverables

- university_raw_data.csv
- data_collection.py

Evaluation

- Successful dataset integration.
 - Dataset completeness above 95%.
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Module 2: Data Cleaning & Transformation

Tasks

- Remove duplicates.
- Standardize university names.
- Standardize country names.
- Normalize ranking metrics.
- Create Tableau-ready datasets.

Deliverables

- university_cleaned.csv
- education_cleaning.ipynb

Evaluation

- Less than 2% missing values.

- Consistent ranking indicators.
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Milestone 2: KPI Engineering and Dashboard Planning (Weeks 3–4)

Module 3: Education KPI Engineering

Tasks

Calculate:

- Global Ranking Score
- Research Impact Score
- Faculty-to-Student Ratio
- International Student Percentage
- Academic Reputation Score
- Research Productivity Index

Deliverables

- university_final_dataset.xlsx
- generate_education_kpis.py

Evaluation

- All KPIs correctly calculated.
 - Dataset optimized for Tableau.
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Module 4: Dashboard Planning & Prototyping

Tasks

Design layouts for:

- University Overview
- Research Analytics

- Student Analytics
- Country Comparison

Define:

- Filters
- Navigation
- Dashboard actions
- Interactive comparisons

Deliverables

- dashboard_storyboard.pdf
- eduvision_prototype.twbx

Evaluation

- Dashboard designs approved.
- Prototype functionality verified.

Milestone 3: Dashboard Development (Weeks 5–6)

Module 5: Build University Overview & Research Analytics Dashboards

Tasks

University Overview

- Top university rankings
- Global university distribution
- Academic reputation analysis
- University performance trends
- Institutional comparison

Research Analytics

- Publications analysis
- Citation performance
- Research productivity trends
- Top research institutions
- Research impact comparison

Deliverables

- eduvision_dashboard_v1.twbx

Evaluation

- Interactive visualizations operational.
 - Ranking metrics validated.
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Module 6: Build Student Analytics & Country Comparison Dashboards

Tasks

Student Analytics

- International student analysis
- Faculty-to-student ratio analysis
- Student diversity trends
- Enrollment comparisons
- Student distribution analysis

Country Comparison

- Country ranking comparison
- Education performance benchmarking
- Regional education trends

- Top performing countries

Dashboard Integration

- Global filters
- Navigation controls
- Parameter actions
- Dashboard linking

Deliverables

- EduVision_DV.twbx

Evaluation

- All dashboards integrated.
 - Navigation and filters functioning correctly.
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Milestone 4: Testing and Delivery (Weeks 7–8)

Module 7: Testing and Validation

Tasks

- Validate KPI calculations.
- Verify ranking calculations.
- Test dashboard interactions.
- Validate educational metrics.

Deliverables

- QA Checklist
- Dashboard Testing Report

Evaluation

- No major dashboard issues.
 - KPI accuracy above 95%.
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Module 8: Documentation and Project Delivery

Tasks

Prepare project documentation including:

- Dataset sources
- KPI definitions
- Dashboard guide
- Education analytics methodology

Organize project structure:

```
/scripts  
/data  
/dashboard  
/docs
```

Deploy:

- GitHub Repository
- Tableau Public (Optional)

Deliverables

- GitHub Repository
- Final Documentation
- Tableau Workbook

Evaluation

- Fully documented project.
 - Portfolio-ready dashboard suite.
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Evaluation Criteria

Milestone	Focus Area	Metric	Target
1	Data Collection & Cleaning	Dataset Completeness	>95% Complete
2	KPI Engineering	KPI Accuracy	6+ KPIs Generated
3	Dashboard Development	Dashboard Functionality	4 Dashboards Integrated
4	Documentation & Delivery	Project Quality	Portfolio Ready

Tech Stack

Area	Tools / Libraries
Data Collection	Python, QS Rankings, World University Rankings
Data Processing	Pandas, NumPy
Data Cleaning	Python
Visualization	Tableau Desktop / Tableau Public
Dashboard Integration	Tableau Filters, Parameters, Actions
Documentation	Markdown, GitHub

Sample Dashboard Layout:



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